Computers in Human Behavior 79 (2018) 30–39

Contents lists available at ScienceDirect

Computers in Human Behavior

journal homepage: www.elsevier.com/locate/comphumbeh

Full length article

Is prosocial video game exposure related to prosociality? An ERP study based on a prosocial help needed decision task

Zhaojun Teng [a, b, 2], Qian Nie [b, 2], Yanling Liu [a, b], Cheng Guo [a, b, *, 1]

[a] The Lab of Mental Health and Social Adaptation, Faculty of Psychology, Southwest University, Chongqing, China
[b] Research Center of Mental Health Education, Southwest University, Chongqing, China

ARTICLE INFO

Article history:
Received 11 April 2017
Received in revised form
26 September 2017
Accepted 12 October 2017
Available online 13 October 2017

Keywords:
Prosocial video game
Prosociality
Event-related potential
N2
P300

ABSTRACT

Previous research has shown that exposure to prosocial video games increases empathy and related-prosocial behaviors. However, little is known about the neural correlates of this effect. This work examined the electrophysiological features of the prosocial video game effect on prosociality by using a help-needed decision task. Seventy participants played either prosocial or neutral video games for 20 min and then participated in an event-related potential (ERP) experiment that estimated the perceived need for help in others and prosocial behavior (i.e., charitable money donated). The results showed no significant differences between the two games in perceived need for help and money donation (prosocial behavior). Moreover, there were inconsistent results for N2 and P300 waves, in which weak evidence was found for larger frontal N2 when help was needed but not offered in the prosocial video game group; however, there were no differences in right-parietal P300 amplitudes. These findings provide no supportive evidence for the general learning model, which proposes that prosocial video games may promote empathy and prosocial-related behaviors. However, we should note the limitation of the small sample size. Further implications and limitations are discussed.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

A dramatic change occurring with the advent of the digital age is that mass media and video games are prevalent in daily life. They have a strong effect on our daily lives. Past research was always concerned about identifying the negative effects of media violence accompanying increasing sales of violent video games (Entertainment Software Association, 2013). Although some researchers have found that exposure to violent video games promotes aggressive behaviors (e.g., Anderson et al., 2010), these studies have been criticized for their publication bias. Recent re-analyses have challenged the aggressive view by suggesting no links between violent video games and adverse outcomes in players (e.g., Ferguson, 2015; Furuya-Kanamori & Doi, 2016; Hilgard, Engelhardt, & Rouder, 2017). To date, there is an ongoing debate regarding the effects of playing violent video games on aggression.

Some researchers, however, have found that prosocial media might increase positive social outcomes (e.g., Gentile et al., 2009; Greitemeyer & Mügge, 2014; Greitemeyer & Osswald, 2010); but these effects were not replicated in other studies (e.g., Tear & Nielsen, 2013). Moreover, a recent electrophysiological investigation suggested that prosocial video games played a critical role in the reduction of aggression through effects on aggressive cognition (i.e., P300 elicited by violent words) (Liu, Teng, Lan, Zhang, & Yao, 2015). However, it is unclear if exposure to prosocial video games has other potential positive effects, in particular for prosocial cognition/empathy. Furthermore, if prosocial cognition/empathy is also affected by prosocial video games, why does this positive effect occur, and what are the underlying neural correlates? This study aimed to address these questions by examining the short-term effects of prosocial video games on a prosocial help needed decision task and characterizing their neural correlates.

1.1. Theory concerning the effect of prosocial video games on prosociality

Although the effects of violent video game have been a focal point in the communication field, some researchers have focused

* Corresponding author.
E-mail address: guochen@swu.edu.cn (C. Guo).
[1] Present address: No 2, Tiansheng Road, Beibei District of Chongqing, China.
[2] The first and second authors have contributed equally to this work.

https://doi.org/10.1016/j.chb.2017.10.014
0747-5632/© 2017 Elsevier Ltd. All rights reserved.

on how prosocial video game exposure might have positive effects. For example, recent research has indicated that exposure to prosocial video games might not only increase prosocial thinking (e.g., Gentile et al., 2009; Greitemeyer & Osswald, 2010, 2011), but also promote empathy as well as prosocial cognition and behavior (e.g., Gentile et al., 2009; Greitemeyer & Osswald, 2010; Greitemeyer, Osswald, & Brauer, 2010; Prot et al., 2014). However, other research has failed to suggest that prosocial video games influence prosociality (e.g., Jerabeck & Ferguson, 2013; Tear & Nielsen, 2013). The general learning model (GLM; Buckley & Anderson, 2006), which is derived from the general aggression model (GAM; Anderson & Bushman, 2002), was often used as the framework in the aforementioned positive studies. The GLM posits that exposure to prosocial media increases empathy and the likelihood of prosocial behavior (Gentile et al., 2009; Greitemeyer & Cox, 2013; Greitemeyer & Osswald, 2010; Prot et al., 2014). If exposure to prosocial video games promotes prosociality (e.g., prosocial thinking, cognition, affect, behavior), it is important to clarify why and when it is most likely to occur.

1.2. Prosocial video game effect on empathy and prosociality

Empathy is the ability to feel and share another person's emotions and thoughts (Singer & Lamm, 2009) and is closely related to prosocial behavior, which is voluntary and intended to benefit another (Eisenberg & Fabes, 1998). Perspective taking, which involves an attempt to understand others' views and feelings (Singer & Lamm, 2009), and empathy for others' pain, which implies use of perception, judgment, and affective responses (Danziger, Prkachin, & Willer, 2006), are two approaches to studying cognitive and emotional empathy, respectively. Basic brain circuits associated with empathy delineate an emotional “low road” and a cognitive “high road” to empathy, which reflect the basic features of emotional and cognitive categorization (Engen & Singer, 2013; Shamay-Tsoory, Aharon-Peretz, & Perry, 2009; Walter, 2012). Even though emotional and cognitive empathy are correlated with prosociality, cognitive empathy may occur readily in daily life events.

Compared to individuals with lower cognitive empathy, those with higher cognitive empathy may find it easier to take others' perspectives. In an intuitive-decision donation task, for instance, people with higher levels of empathy made prosocial decisions more quickly and donated a larger amount of money than did those with low empathy (Carlson, Aknin, & Liotti, 2015). Empathy is closely associated with prosociality. In a task in which help was needed urgently (e.g., because of injury or hemorrhage), the decision to help was always made more quickly than it was in neutral conditions, perhaps because this situation activated individuals' prosocial scripts (Chiu Loke, Evans, & Lee, 2011). Perceived need for help in others and the decision to help quickly might be affected by contextual variables such as video game exposure (e.g., Greitemeyer & Osswald, 2010). Furthermore, empathy might mediate the effect of prosocial video game exposure on prosociality (e.g., Greitemeyer et al., 2010; Prot et al., 2014). Therefore, if playing prosocial video games helps increase empathy, people might recognize the need for help in others more quickly in conditions involving urgent situations (e.g., injury or hemorrhage).

1.3. Neural correlations of prosociality

Even though few studies have examined empathy and prosociality using event-related potentials (ERPs), the high temporal precision of ERPs is well suited to revealing the rapid neural reactions associated with cognitive empathy that might resulting from playing prosocial video games. In particular, the parietal P300, which is a positive ERP component peaking roughly 300–400 ms after a related event, is associated with prosociality and empathy. Recent research has shown increases in P300 amplitudes in evaluative decision-making processes involving the locus coeruleus norepinephrine system when motivational salience was rapidly encoded (Hajcak, MacNamara, & Olvet, 2010; Nieuwenhuis, Aston-Jones, & Cohen, 2005; Polich, 2007). Moreover, the P300 component appears to be sensitive to event stimuli activating the reward motivational system in the mesolimbic regions (Carlson et al., 2015; Cox et al., 2015). When stimuli lead to greater arousal, attention, or perception, the reward motivation system shows greater activity (i.e., larger P300). Moreover, a larger P300 is positively related to empathy (Fan & Han, 2008; Luo et al., 2014), prosocial personality (Chiu Loke et al., 2011), prosocial attitudes (Xiao, Zheng, Wang, Cui, & Chen, 2015), and donating behaviors (Carlson et al., 2015).

Another important component relevant to the study of prosociality is the frontal N2, which displays negative deflections over frontal sites between 180 and 380 ms after stimulus onset or feedback. This component has reliably been used as an index of cognitive conflict detection (e.g., Folstein & Petten, 2008), and may be localized to the anterior cingulate cortex (e.g., Hajcak, Moser, Yeung, & Simons, 2005; Van Veen & Carter, 2002), a region implicated in conflict monitoring. Largely negatively N2 amplitudes seem to occur when conflict is high, such as when an inconsistent word–color match appears in a color-word Stroop task (Szűcs & Soltész, 2010). Moreover, it has been suggested that larger N2 amplitudes occur when a rule was removed rather than assumed for moral, but not conventional, violations (Lahat, Helwig, & Zelazo, 2013). The N2 is also considered a marker of emotion in prosocial behavior, as its amplitude shows a significant positive relationship with high-empathic profiles (e.g., Balconi & Canavesio, 2013).

Recent studies examining prosociality and empathy for others' pain suggested that N2 and P300 amplitudes reflected the ability to evaluate pain from another's perspective, over the frontal and centroparietal regions associated with empathic responses (Fan & Han, 2008; Luo et al., 2014; Yoder & Decety, 2014). Moreover, larger P300 amplitudes were observed over centroparietal regions in response to high-empathy, rather than low-empathy, targets (Carlson et al., 2015). Additionally, one study found larger N2 and P300 amplitudes in response to implicit prosocial attitudes in a conflict control situation, as well as a positive relationship between N2 amplitude differences and the amount of money donated (Xiao et al., 2015). Also, the ability to perceive others' need for help was associated with larger P300 amplitudes in response to decisions not to help when help was needed (i.e., a conflict situation), and that the P300 was positively correlated with prosociality (Chiu Loke et al., 2011). Overall, these above studies revealed that ERP components (i.e., the N2 and P300) are closely associated with prosociality and empathy. However, are these ERP characteristics also affected by contextual variables, such as prosocial video game exposure? The present study aimed to answer this question.

Previous studies have examined both the short- and long-term effects of prosocial video games on empathy and prosociality (e.g., Greitemeyer & Osswald, 2010; Prot et al., 2014). In behavioral terms, according to the GLM, exposure to prosocial video games might increase prosociality (e.g., Gentile et al., 2009; Greitemeyer & Osswald, 2010; Greitemeyer et al., 2010; Prot et al., 2014), although some researchers have found no support for this (e.g., Jerabeck & Ferguson, 2013; Tear & Nielsen, 2013). However, there are still limited data available regarding the neural correlates of the effect of prosocial video game exposure on social outcomes. In a recent ERP study, Liu et al. (2015) found lower levels of P300 activity in response to violent words following short-term exposure to a prosocial video game, and provided new evidence indicating that prosocial video game exposure could lower aggression via

reduction in the mediator of aggressive cognition (i.e., lower levels of P300). However, although their results are encouraging, they did not control for the game experience of participants. Moreover, Liu et al. (2015) did not consider the effects of prosocial video game exposure on prosociality. The present study collected data concerning each participant's game-playing experience and explored the promotional effect of playing a prosocial video game on helping decisions using the ERP method.

1.4. The present study

The present study included two conditions (urgent need for help vs. neutral) in which stories were presented to participants according to Chiu Loke et al. (2011) paradigm. First, we matched two video games, one prosocial and one non-prosocial, for use in the formal experiment. Following exposure to either a prosocial or a non-prosocial video game for 20 min, participants completed an ERP task to measure their perception of others' need for help using the help-needed decision task (Chiu Loke et al., 2011). Finally, participants' willingness to give a charitable donation was assessed as a measure of prosociality.

Based on the background above, we predicted that N2 and P300 amplitudes would become larger after short-term exposure to a prosocial video game. Specifically, in conditions involving the perception of others' urgent need for help but wherein help was not offered (i.e., a conflict situation), participants who had played prosocial video games were expected to produce larger P300 amplitudes in the right-parietal regions, and more negative N2 peak amplitudes in the frontal regions, relative to those who had played a non-prosocial video game. As to the charitable donation, we also predicted there would be higher donations in the prosocial video group than in the non-prosocial video game group.

Recently, there have been many critiques of reporting ERP experiments and interpreting ERP results. In response, detailed coverage of methods is presented, including how we measured the ERPs, sample size decisions, and invalid data exclusions. For null hypothesis testing, we report detailed *p*-values, standardized effect-size estimates (Cohen's *d*), power, and 95% confidence intervals (*CI*s) in the results section. To assess the degree of confidence in the results, we also performed a Bayes factor analysis of the data.

2. Materials and methods

2.1. Participants

Participants included 70 right-handed students (37 female, mean age = 21.50 years, SD = 1.88) from Southwest University, China. Participation in the study was voluntary, and all participants provided informed consent. All participants in the formal experiment had normal or corrected-to-normal vision, and none had taken any medication recently or suffered from neurological defects or damage (e.g., language impairment). No participants guessed the aim of the study, and none majored in a related subject (e.g., psychology).

The sample size of both the prosocial video game group and non-prosocial video game group (i.e., control group) was set to 35 (19 and 15 females, respectively). This stopping rule (35 per group) was determined prior to random recruiting of participants because of limited funding and based on the notion that ERP studies usually have 20 participants in each group. This study was approved by the Human Research Ethics Committee of the Southwest University of China. Upon completion of the study, participants were paid RMB 40 and thanked for their time.

2.2. Materials

2.2.1. Video games

Two video games were tested in a preliminary study: *Lemmings*, a prosocial video game widely used in previous research (Gentile et al., 2009; Greitemeyer & Cox, 2013; Greitemeyer & Osswald, 2010), and *Ball-Buster Collection*, a non-prosocial video game. In *Lemmings*, players guide groups of small beings through different worlds. The goal is to take care of the beings and save them by leading them to the exit. In *Ball-Buster Collection,* players must control the mouse and make a ball bounce so that it does not fall to the ground. All participants played the games on the same computer, and needed to operate the mouse to carry out the game task.

Participants were evaluated following completion of the games. They completed the Game Evaluation Questionnaire (GEQ; Anderson & Dill, 2000), in which they rated each game on six dimensions (frustration, enjoyment, difficulty, proficiency, prosocial and aggressive content, and actions) using a 5-point scale ranging from 1 to 5. Participants' emotional state was measured using the Positive and Negative Affect Schedule (PANAS; Watson, Clark, & Tellegen, 1988), which consists of 10 positive and 10 negative mood items; participants rated the extent to which they experienced each of the emotions described, after playing video games, using a 5-point scale ranging from 1 to 5. Current arousal was evaluated using the Perceived Arousal Scale (PAS; Anderson, Deuser, & Deneve, 1995), which contains a list of 31 adjectives describing feelings of arousal or lack of arousal; participants indicate the extent to which the adjectives describe their feelings using a 5-point scale ranging from 1 to 5.

We performed a preliminary game evaluation of 50 participants (38 women) who were not part of the formal experiment. Each participant completed two video game evaluations. First, they played one of the games for 3 min in a practice round, to ensure that they understood the rules. Participants then played the games for 20 min (official round), at which point, the experimenter stopped the game and asked them to complete the Game Evaluation Questionnaire, the PANAS, and the PAS. After a break lasting 3 min, participants followed the same procedure to evaluate the other game.

2.2.2. Video game experience questionnaire

The Video Game Questionnaire, developed by Anderson and Dill (2000), was used to measure video game exposure. As in a study conducted by Prot et al. (2014), participants were instructed to list their three favorite video games involving prosociality. After listing the name of each game, they were asked to record the frequency with which they played each game (1 = rarely, 7 = often), and the prosociality of the game content (1 = a little, 7 = extremely prosocial). The average rating for the three video games listed by each participant was used as an overall index of exposure to video games (Anderson & Dill, 2000; Prot et al., 2014). We calculated a prosocial exposure score for each of the participants' three games by averaging the prosociality ratings and multiplying this by the previously calculated frequency score.

2.3. Experimental tasks

We designed two types of story: in one, another person is described as needing help urgently (e.g., “David is walking down the road and sees someone who has fallen and has blood dripping down his arm. David decides to …”), while in the other, a neutral event in which help was not required is described (e.g., “Jim is out for a walk and sees someone trying to take off his rollerblades. Jim decides to …”). We provided two outcome decisions “help” or “not help” (e.g., David decides “help”). Thus, consistent (i.e., help/need-

help and no help/need-no help) and conflict events (i.e., help/need-no help and no help/need-help) were contained in the above four situations.

The stories were designed to be similar in length and matched in content across situation types and helping actions (in Chinese). Each story was presented in a short segment in the center of a computer screen, using E-Prime 2.0 (Psychological Software Tools, Pittsburgh, PA). A task trial is presented in Fig. 1. Participants were allowed 2500 ms to read the story, which gave them enough time to clearly comprehend the story, after which a black screen was shown randomly for 500–1050 ms. Thereafter, either "help offered" or "help not offered" appeared randomly on the screen for 1000 ms. Participants then assessed the decision of the person in the story, choosing "good" or "bad" by pressing the "f" or "j" key, respectively. However, the "good" or "bad" decision held no special meaning, which simply allowed the participants to remain focused on the stories.

2.4. Experimental design and procedure

Participants were advised that they would take part in an EEG/ERP experiment to assess the effect of video games on story reasoning. First, participants were guided by the trained researcher to grasp the game operation, which took an average of about 6 min. After the participant understood the game rules and operation, the game practice ended.

Second, the EEG helmet was fitted to the participant and electrode jelly applied. Subsequently, participants were randomly assigned to either the prosocial- or neutral-game group and played the allocated game for 20 min. This 20 min allowed us to assess the short-term effects of prosocial video games, as in previous research (Greitemeyer & Osswald, 2010; Liu et al., 2015). Following a 2-min break, participants were asked to complete a help-needed decision task, which involved assessing decisions within a social context. The training trials for this task lasted 5 min. Thereafter, the formal experiment was initiated.

There were a total of 60 stories, including 30 help-needed-urgently stories and 30 help-not-needed-urgently (i.e., neutral) stories. For each story, there were two versions wherein the protagonist decided to help or not help (for a total of 120 story-decision combinations, equating to 120 trials). We repeated each story-decision combination twice, so that participants were exposed to a total of 240 trials. After the help-needed decision task, participants were asked to rate their willingness to give a charitable donation ("Yes" or "No") for pupils in poor mountainous areas. If they answered "yes," they were asked how much money they would donate, which acted as a behavioral measure of prosociality.

2.5. EEG recording and data processing

Continuous EEG recordings were taken using 64 Ag-AgCl unipolar leads on a 64-lead connection, consistent with the extended 10–20 system, and digitized at a sampling rate of 500 Hz (Brain Products GmbH). The impedance for all electrodes was kept below 5 kΩ. The prefrontal electrode, at the mid-point between FPz and Fz (i.e., AFz), was connected to the ground with the frontal vertex (i.e., FCz) as the online reference. Participants were asked to refrain from blinking, remain still, and relax their facial muscles. They were required to look at a fixation cross, which was presented on the screen. To control for eye movement artifacts, horizontal and vertical electrooculograms were recorded from electrodes above and at the outer canthus of the right eye, respectively.

Offline analysis of the EEG data was performed using Analyzer 2.0 software (Brain Products GmbH), with recordings re-referenced to the "infinity" reference provided by the reference electrode standardization technique (REST; Yao, 2001). Regression was performed to remove the horizontal and vertical electrooculograms (EOGs) from the Analyzer 2.0 system, using a 0.01–30 Hz band pass filter and a corrected baseline of −200 ms in Analyzer 2.0. Trials with EOG artifacts (mean EOG voltage exceeding±80 μV) and those contaminated with artifacts because of amplifier clipping, bursts of electromyography activity, or peak-to-peak deflection exceeding ±80 μV were excluded from averaging. Table 3 displays the resting mean trials for each condition.

According to previous research (Balconi & Canavesio, 2013; Lahat et al., 2013; Xiao et al., 2015; Yoder & Decety, 2014), we analyzed the average peak amplitudes of spatially related N2 in the frontal area (F3, Fz, and F4). The peak detection time windows for N2 were 180–380 ms. Because the P300 component has been observed predominantly in the parietal regions (Carlson et al.,

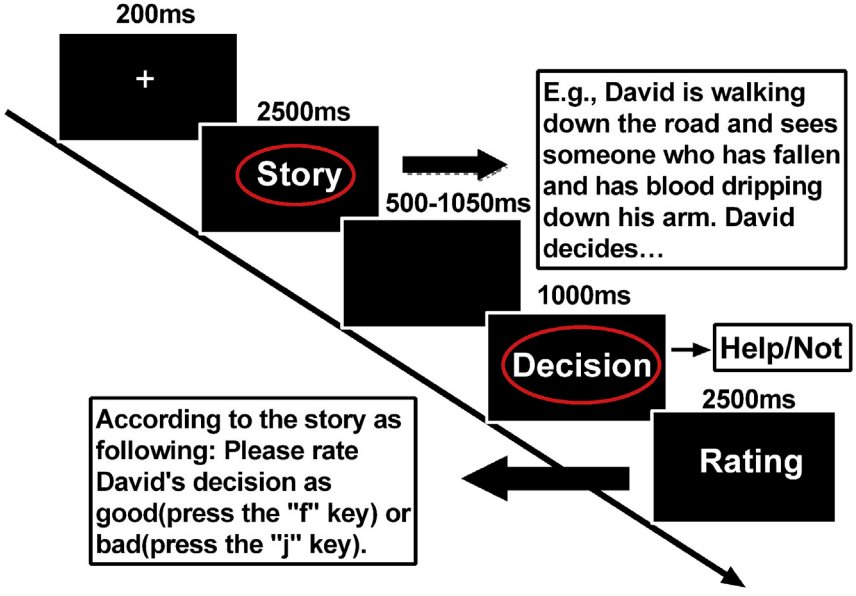


Fig. 1. Prosocial help-decision task in one trial.

2015; Fan & Han, 2008), and P300 effects are particularly evident in the right-parietal regions for prosocial help needed task (Chiu Loke et al., 2011), we analyzed average amplitudes of spatially related P300 in the right-parietal area (CP4, P4, and PO4). The time windows for P300 waves were 250–500 ms with average amplitudes (Luck, 2014). An epoch of −200 to 1000 ms and a baseline of 200 ms prior to stimulus onset were established for each trial. Grand average N2 and P300 amplitudes were calculated for F3/Fz/F4 in the frontal and CP4/P4/PO4 in the right-parietal area (see Fig. 2).

A 2 (game type: prosocial and non-prosocial) × 2 (condition: help needed and neutral) × 2 (decision: help offered and help not offered) repeated measures analysis of variance (ANOVA) was performed to examine reaction times (RTs) in the behavioral task, as well as the average N2 and P300 amplitudes in the ERP data. To measure effect sizes, Cohen's *d* and partial eta-squared (η^2_p) values (Cohen, 1988) were calculated for all significant *t* or *F* values; 95% CIs were also provided for manipulation check data. The 95% CIs were derived using approximations for the noncentral *t* distribution (see Algina & Keselman, 2003). A statistical power (1-β) was also calculated for each effect. Moreover, to enhance reliability, the data were analyzed using a Bayesian version of the repeated-measures ANOVA in R (e.g., Morey, Rouder, & Jamil, 2014). Three models were computed by the Bayes factors analysis: (1) the full model of the interactions and the main effects of game type, conditions, and decisions; (2) a model with a main effect but without any interactions; and (3) a null model. $BF_{1,0}$ has been suggested to indicate the level of confidence to support the alternative hypothesis as compared with the null hypothesis. For example, $BF_{1,0=}10$ means that the odds are 10 to 1 in favor of the alternative hypothesis compared to the null hypothesis. Bayes factors were calculated by using the default prior (*r* scale = 0.5, c.f., Rouder, Morey, Speckman, & Province, 2012) for all ANOVAs of RTs, N2, and P300.

3. Results

3.1. Manipulation check

3.1.1. Video games

One participant wrote the same answer for all measurements, so their data were excluded; hence, 49 participants (37 females) were included in the analysis (the data see S1 file). A paired-samples *t*-test was performed to compare the two games (see Table 1). There were no significant main effects of affect (e.g., PA and NA), arousal, frustration, enjoyment, difficulty, proficiency, aggressive content, or action (all $p > 0.10$). However, a significant main effect of prosocial perspective (e.g., content and action) was observed ($p < 0.01$), with higher levels of prosocial content and action in the prosocial video game relative to those in the non-prosocial game. This indicated a successful distinction between the prosocial and non-prosocial environments in the two games, and they were well matched for other related variables.

3.1.2. Video game experience

There was no difference in video game experience, $t_{(66)} = -0.61$, $p = 0.55$, $d = -0.15$, 95%*CI* [−3.24, 1.73], between the prosocial ($M = 11.86$, $SD = 5.25$) and non-prosocial video game groups ($M = 12.62$, $SD = 5.02$).

3.2. Behavioral results

3.2.1. Reaction time

Concerning the prosocial help-decision task, the decision of "good" or "bad" was selected quickly in the response phase. The RTs reflected participants' ability to estimate the story in line with their moral standards. However, RT had no special meaning in the present study, as this was simply a way to keep the participants focused on the story. No correct or error responses were provided for participants who selected "good" or "bad." Therefore, only RTs were included in the data analysis. According to the principles of analyzing RTs, all trial RTs that were greater than 200 but no more than 3000 were included in the statistics for mean RTs (for the data, see S2 file).

Table 2 displays the statistics for the RTs. There were significant main effects of condition $F_{(1,66)} = 30.66$, $p < 0.01$, $\eta^2_p = 0.32$, power $(1-\beta) > 0.99$, $BF_{1,0} = 469.70$, and decision, $F_{(1,66)} = 85.05$, $p < 0.01$, $\eta^2_p = 0.56$, power $(1-\beta) > 0.99$, $BF_{1,0} = 1.61 \times 10^{15}$, with shorter RTs in the help-needed condition ($M = 770.71$, $SE = 22.19$) than in the neutral condition ($M = 859.72$, $SE = 25.99$). RTs were also shorter for the help-offered decision ($M = 724.20$, $SE = 20.78$) than for the help-not-offered decision ($M = 906.23$, $SE = 28.32$). However, there was no main effect of game type, $F_{(1,66)} = 0.74$, $p = 0.388$, $\eta^2_p = 0.01$, power $(1-\beta) = 0.14$, $BF_{1,0} = 0.24$. No two-way or three-way interaction effects were found (all $F < 1.50$, $p > 0.23$, $\eta^2_p < 0.03$, $BF_{1,0} < 0.65$).

3.2.2. Donation

A total of 16 participants chose to donate no money, and there was no difference between the prosocial ($n = 6$) and non-prosocial video game groups ($n = 10$), $\chi^2 = 1.31$, $p = 0.253$, $BF_{1,0} = 0.64$ (prior concentration is 1) in this decision. Fifty-two participants chose to donate money, and there was also no difference, $t_{(50)} = 0.11$, $p = 0.916$, Cohen's $d = 0.03$, power $(1-\beta) = 0.05$, $BF_{1,0} = 0.37$ (prior scale is 0.5), between the prosocial ($M = 20.71$, $SD = 10.43$) and non-prosocial video game groups ($M = 20.42$, $SD = 9.66$). Further, no difference [$t_{(66)} = 0.88$, $p = 0.382$, Cohen's $d = 0.21$, power $(1-\beta) = 0.14$, $BF_{1,0} = 0.44$ (prior scale is 0.5)] was found between the prosocial ($M = 17.06$, $SD = 12.12$) and non-prosocial video game groups ($M = 14.41$, $SD = 12.13$) when considering all participants (when treating the non-donations as zero).

3.3. ERP results

For the ERP data, we excluded the data for average trials less than half of the whole trials (i.e., less than 30) after artifact exclusion. We excluded 2 participants (one in the prosocial video game group and another in the control group) (for the data see S2 file). All valid trial means were near about 55 (see Table 3). Grand average waveforms elicited by the stories for every condition in the two game groups are shown in Fig. 2.

3.3.1. N2 (180–380 ms)

For N2 peak amplitudes, there were significant main effects of condition and decision, $F_{(1,66)} = 6.88$, $p = 0.01$, $\eta^2_p = 0.09$, power $(1-\beta) = 0.73$, $BF_{1,0} = 3.25$, and $F_{(1,66)} = 28.74$, $p < 0.01$, $\eta^2_p = 0.30$, power $(1-\beta) > 0.99$, $BF_{1,0} = 1.79 \times 10^5$, respectively, with more negative amplitudes for N2 in the help needed condition ($M = -3.42$, $SE = 0.28$) than in the neutral condition ($M = -3.07$, $SE = 0.25$). N2 peak amplitudes were also more negative for help not offered ($M = -3.60$, $SE = 0.28$) than for help offered decisions ($M = -2.89$, $SE = 0.25$). However, there was no main effect of game type, $F_{(1,66)} = 0.28$, $p = 0.599$, $\eta^2_p < 0.01$, power $(1-\beta) = 0.08$, $BF_{1,0} = 0.25$. There were also no two-way interactions of condition × decision, condition × game type, decision × game type (all, $F < 0.50$, $p > 0.45$, $\eta^2_p < 0.01$, $BF_{1,0} < 0.30$). However, there was a three-way interaction of condition × decision × game type, $F_{(1,66)} = 6.09$, $p = 0.016$, $\eta^2_p = 0.09$, power $(1-\beta) = 0.68$, $BF_{1,0} = 1.45$. A simple effects test was conducted (Fig. 3). For the prosocial video game group, there was no difference between the help needed and neutral conditions with the decision of help offered, $F_{(1,66)} = 0.38$,

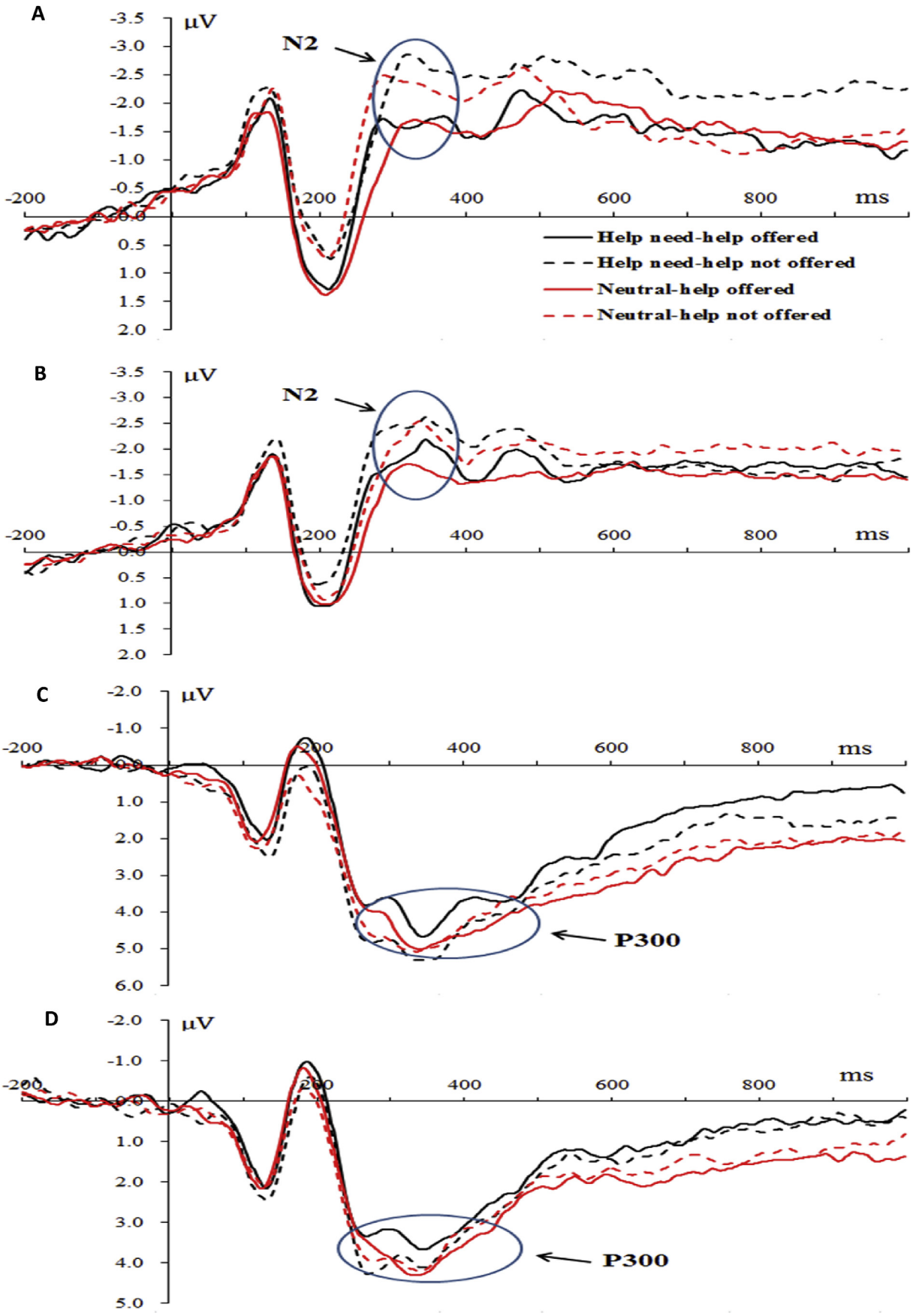


Fig. 2. Grand average waves of N2 and P300. (A) The frontal N2 waves for the prosocial video game group, (B) the frontal N2 for the neutral video game group, (C) the right-parietal P300 for the prosocial video game group, (D) the right-parietal P300 for the neutral video game group.

Table 1
ANOVA results for each dimension of the video games ($n = 49$).

Dimension	*Lemmings*	*Ball-Buster Collection*	$t_{(48)}$	p	d
Positive affect	3.36 (0.65)	3.49 (0.61)	−1.27	0.210	−0.13 [0.08, −0.35]
Negative affect	1.61 (0.58)	1.49 (0.40)	1.52	0.136	0.12 [−0.04, 0.28]
Arousal	2.38 (0.30)	2.43 (0.27)	−1.66	0.104	−0.05 [−0.12, 0.02]
Frustration	2.84 (1.07)	2.59 (1.12)	1.25	0.219	0.24 [−0.15, 0.64]
Enjoyment	4.14 (0.76)	4.24 (0.63)	−0.82	0.417	−0.10 [−0.35, 0.15]
Difficulty	2.78 (1.04)	2.49 (0.96)	1.37	0.176	0.29 [−0.13, 0.70]
Proficiency	3.49 (0.89)	3.69 (0.87)	−1.22	0.229	−0.20 [−0.54, 0.13]
Prosocial content	3.76 (0.94)	2.04 (0.84)	8.97	<0.001	1.71 [1.33, 2.10]
Aggressive content	1.67 (0.83)	1.61 (0.61)	0.47	0.644	0.06 [−0.20, 0.32]
Perceived prosocial action	3.22 (1.01)	1.59 (0.67)	9.51	<0.001	1.63 [1.29, 1.98]
Perceived aggressive action	1.29 (0.82)	1.22 (0.55)	0.54	0.595	0.06 [−0.17, 0.29]
Violent scenes	1.94 (0.85)	1.88 (0.78)	0.42	0.679	0.06 [−0.23, 0.36]

Table 2
Reaction times for four conditions (M ± SD; ms).

	Prosocial video game		Non-prosocial video game	
	Help needed	Neutral	Help needed	Neutral
Help offered	670 ± 178	745 ± 187	672 ± 179	809 ± 197
Help not offered	851 ± 270	915 ± 263	889 ± 192	969 ± 286

Table 3
Describe of the valid trials for four conditions (M ± SD).

	Prosocial video game		Non-prosocial video game	
	Help needed	Neutral	Help needed	Neutral
Help offered	55.71 ± 6.06	55.85 ± 6.38	56.29 ± 5.14	55.85 ± 5.86
Help not offered	56.00 ± 6.61	55.50 ± 6.02	55.50 ± 6.96	55.79 ± 6.76

$p = 0.539$, $\eta^2_p < 0.01$, power $(1-\beta) = 0.09$, $BF_{1,0} = 0.22$, however, there were more negative N2 peak amplitudes for the neutral condition than for the help needed condition with the decision of help not offered, $F_{(1,66)} = 4.85$, $p = 0.031$, $\eta^2_p = 0.07$, power $(1-\beta) = 0.58$, $BF_{1,0} = 1.02$. For the non-prosocial video game group, there were more negative N2 peak amplitudes for the help needed than for the neutral condition with the decision of help offered, $F_{(1,66)} = 8.10$, $p = 0.006$, $\eta^2_p = 0.11$, power $(1-\beta) = 0.80$, $BF_{1,0} = 4.12$; however, there were no differences between the help needed and neutral conditions with the decision of help not offered, $F_{(1,66)} = 0.06$, $p = 0.807$, $\eta^2_p < 0.01$, power $(1-\beta) = 0.06$, $BF_{1,0} = 0.19$. For N2 peak latency, only a main effect of decision was found, $F_{(1,66)} = 4.28$, $p = 0.042$, $\eta^2_p = 0.06$, power $(1-\beta) = 0.53$, $BF_{1,0} = 0.95$, with an earlier N2 peak latency for the help offered decision than for the help not offered decision.

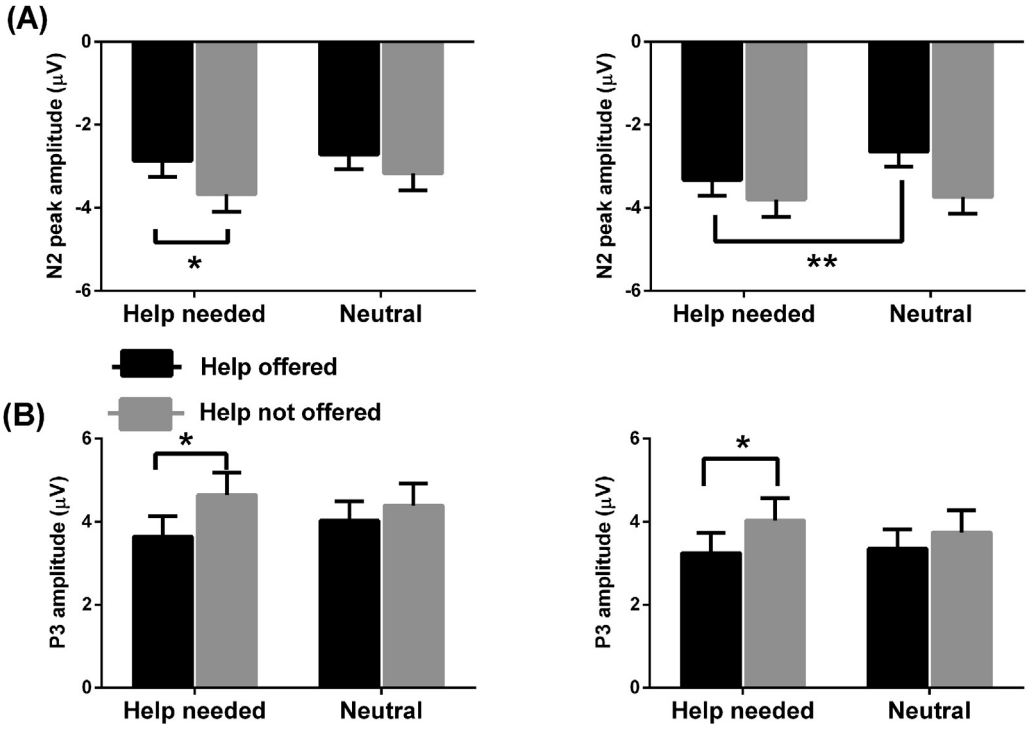


Fig. 3. The average amplitudes (uV). (A) The N2 peak amplitudes, with the left representing the prosocial video game group and the right representing the non-prosocial video game group. (B) The P300 amplitudes, with the left representing the prosocial video game group and the right representing the non-prosocial video game group. $^{*}p < 0.05$; $^{**}p < 0.01$, Error bars stand for ±1 SE.

3.3.2. P300 (250–500 ms)

There was a main effect of decision, $F_{(1, 66)} = 17.11$, $p < 0.001$, $\eta^2_p = 0.21$, power $(1-\beta) = 0.98$, $BF_{1,0} = 1.31 \times 10^4$, with the help not offered decision presenting larger P300 amplitudes ($M = 4.20$, $SE = 0.38$) than the help offered decision ($M = 3.57$, $SE = 0.32$). No main effects of condition or game type were observed (all $F < 1.00$, $p > 0.39$, $\eta^2_p < 0.01$, $BF_{1,0} < 0.32$). However, there was an interaction of condition × decision, $F_{(1, 66)} = 7.96$, $p = 0.006$, $\eta^2_p = 0.11$, power $(1-\beta) = 0.79$, $BF_{1,0} = 1.26$. A simple effects test revealed that the help needed condition presented larger P300 amplitudes than did the neutral condition with the decision of help not offered, $F_{(1, 66)} = 4.58$, $p = 0.036$, $\eta^2_p = 0.07$, power $(1-\beta) = 0.56$, $BF_{1,0} = 1.16$; however, no difference was found between the help needed and neutral conditions with the decision of help offered, $F_{(1, 66)} = 1.89$, $p = 0.174$, $\eta^2_p = 0.03$, power $(1-\beta) = 0.27$, $BF_{1,0} = 0.33$. No other two-way or three-way interactions were found (all $F < 0.50$, $p > 0.51$, $\eta^2_p < 0.01$, $BF_{1,0} < 0.30$).

4. Discussion

The aim of this study was to provide insight into the effect of short-term exposure to prosocial video games on prosociality and its neural correlates via a help-needed decision task. Inconsistent with our hypothesis—namely, that greater prosocial behaviors (i.e., willingness to donate, amount of money donated) and higher N2 and P300 amplitudes would occur in the help needed condition when help was not offered in the prosocial video game group—no significant results for behavioral outcomes (e.g., RTs or donations) or neural correlates (e.g., N2 or P300) were found. This fails to support the behavioral outcomes and neural correlates expected by GLM (Buckley & Anderson, 2006; Gentile et al., 2009), which proposes that prosocial video games can promote empathy or prosociality. The frontal N2 amplitudes were more negative in the help needed but not offered condition following short-term exposure to a prosocial video game. By contrast, there were more negative N2 amplitudes in the non-prosocial video game group in the help offered and help needed condition, but not in the prosocial video game group. This was consistent with previous research that N2 peak amplitudes were more negative in response to prosocial-related conflict emotions and events (Balconi & Canavesio, 2013; Lahat et al., 2013; Xiao et al., 2015; Yoder & Decety, 2014). We did not find any P300 effect in the right-parietal area in the help needed but not offered condition; larger amplitudes were present in both game groups, not only in the prosocial video game group.

First, we observed not only no difference in whether a donation was made (yes or no to donation) between the two game groups, but also no difference in the amount of money donated. This result was inconsistent with Greitemeyer and colleagues' work, which showed significant prosocial behaviors for prosocial video game playing. However, this result accorded with that of Tear and Nielsen (2013), who failed to find evidence that playing video games affects prosocial behavior. It is possible that by the onset of early adulthood (i.e., college students), the factors contributing to prosocial inclinations have been well-established and are not easily shifted by playing a video game for 20 min. Another possible concern is our small sample size ($n = 70$), which limited each game condition to about 35 participants. Second, there were no significant interactions among the three factors (game type, condition, and decision), and there were only main effects of condition and decision for the RTs.

Moreover, consistent with previous research, we found two-way interactions of condition and decision for the P300 in the right-parietal area (Chiu Loke et al., 2011). However, there was no group effect. Specifically, participants had larger P300 activation in response to the help not offered decision when help was obviously needed, and in response to the help offered decision when help was not needed (i.e., the neutral condition). This result only suggests that conflict situations are highly activating and elicit larger P300 amplitudes (Chiu Loke et al., 2011; Xiao et al., 2015). Larger P300 amplitudes have been shown to indicate a cognitive empathic response (Fan & Han, 2008; Luo et al., 2014). This finding differs from that concerning emotional empathy, which appears to elicit larger N1 or P2 amplitudes (Fan & Han, 2008). The results of the present study suggested that the condition involving taking another's perspective to perceive their urgent need for help (e.g., because of injury or hemorrhage) might activate cognitive empathy.

Larger N2 amplitudes were observed in the prosocial game group in response to the perception of others' need for help but only when help was not offered (i.e., a conflict situation). These results are somewhat consistent with those of a previous study (Chiu Loke et al., 2011), in which larger P300 amplitudes (but not N2 amplitudes) were observed for the help-not-offered decision in response to perceiving others' need for help. The Bayes factor also suggested weak evidence of an effect on N2 peak amplitudes in the frontal area, which was also consistent with previous research (Lahat et al., 2013; Xiao et al., 2015), when we set the prior to 0.5. Because of empathic responses to others' perspectives, people will increase their empathy related cognition following prosocial video game play, though the evidence is weak. Therefore, we should be cautious about interpreting this finding.

To our knowledge, this might be the second neural correlation study to explore prosocial video game effects on social behaviors. Liu et al. (2015) first used a prosocial video game as a priming condition to reveal that positive outcomes might have neural correlations, and found that prosocial video games might lower P300 activity elicited in response to violent words. However, their study also found an indirect effect of reducing P300 activity elicited in response to prosocial words (Liu et al., 2015). From our results (Fig. 3), we found no P300 effect, but did find a small trend towards a reduced N2 in response to prosocial video game playing. Thus, we should conduct more studies to clarify whether prosocial video games lead to the reduction or enhancement of certain neural activities.

Considering the P300, prosocial video games—relative to violent video games, which are associated with the activation of aversive motivations (decision-making processes related to withdrawal behavior; Bartholow, Bushman, & Sestir, 2006; Engelhardt, Bartholow, Kerr, & Bushman, 2011)—have been implicated in the activation of reward motivation, which involves decision-making processes related to prosocial behavior. For example, the results of an ERP/fMRI study showed larger P300 amplitude and ventral striatum activity in response to rewarding cues (e.g., Pfabigan et al., 2014), indicating that P300 modulation captured early motivational processes that drive reward-seeking behavior. Another fMRI study found that prosocial motivation (e.g., Morelli, Rameson, & Lieberman, 2014) activated the septal area, which is associated with a variety of prosocial emotions and behaviors and can predict daily prosocial behavior. Moreover, a larger P300 amplitude was observed for high-empathy targets, relative to low-empathy targets, in decision making in intuitive conditions (e.g., Carlson et al., 2015). However, the P300 result did not reach significance in this study.

Nevertheless, further studies should be conducted to determine whether prosocial video games do or do not prime the reward motivation system (i.e., P300) to display prosocial behaviors. The effect of prosocial video games has been consistently studied, but a conclusion has not yet been reached: some studies have supported the prosocial video game effect (e.g., Gentile et al., 2009; Greitemeyer & Osswald, 2010), and some have not (e.g., Jerabeck

& Ferguson, 2013; Tear & Nielsen, 2013). In contrast, the results concerning the neural correlates of this effect are limited. Our results extend previous findings by identifying the neural correlates via a prosocial help decision task following short-term prosocial video game exposure.

However, this study has failed to provide any evidence, whether behavioral or neural, in support of the GLM. While the GLM (Buckley & Anderson, 2006) might provide a useful framework for prosocial media research, its validity has been recently questioned. First, the GLM was basically derived from the GAM, which has received considerable criticism over the years (e.g., Ferguson & Dyck, 2012; Kutner & Olson, 2008) and for which many experiments have failed to find supportive evidence (e.g., Przybylski, Deci, Rigby, & Ryan, 2014; Read, Ballard, Emery, & Bazzini, 2016; Sauer, Drummond, & Nova, 2015). Moreover, recent neuroscience research has also failed to find evidence for neural empathic responses (e.g., Szycik, Mohammadi, Münte, & Te Wildt, 2017b) and emotional desensitization (e.g., Gao et al., 2017; Szycik et al., 2017a), which not supports the GAM. Second, most past research concerned prospective cognitive and emotional effects for prosocial video games; however, little research had been conducted on arousal (Greitemeyer, 2011).

The present study was subject to several limitations. First, although an effort was made to control for cognition, affect, arousal, difficulty, enjoyment, and other attributes of the two types of video game, various other aspects, such as competitiveness, were not. In addition, prosocial games currently available for study, such as *Lemmings*, indirectly involve violence. Therefore, future research should control for key variables as far as possible. There are numerous violent video games available, which are universally popular; however, there are few prosocial video games on the market. Hence, future research should design prosocial video games to be used in research examining the effects of prosocial media further. Second, due to random participant selection, we did not consider prosocial traits or trait empathy, which could be important factors in the present research (e.g., Chiu Loke et al., 2011) and should have been matched between the two groups. Some studies have shown that agreeableness, one of the Big Five Personality traits, is a key predictor of prosocial behavior (Graziano, Habashi, Sheese, & Tobin, 2007). Future studies should control for personality traits. Third, the power in the study should be considered a limitation. As the study only included 70 participants, the statistical power might be weak. Future studies should include more participants (e.g., $n = 150$) to obtain greater power. Fourth, the prior plays a crucial role in the Bayes factors of the ERP data; however, using 0.5 as the prior in the present analysis might have limited the effect size. Thus, further studies should consider a more appropriate prior when using Bayesian analysis. In addition, the present study examined the short-term effects of prosocial video game exposure on cognitive empathy, but the long-term effects are unclear and should be explored in future research.

5. Conclusion

The present study identified the neural correlates of prosocial video game on prosociality, although it is important to note that this is a preliminary study and future work is needed. To clarify whether the GLM has some neural correlates, we conducted an ERP study following short-term exposure to a prosocial video game, in which participants performed a help-needed decision task. However, the results showed no significant behavioral differences (either in RTs or donating money) between the prosocial and non-prosocial video game groups, although this cannot be considered conclusive given the limitation of the small sample size. We did obtain weak significant evidence that a more negative frontal N2 occurs in the help needed but not offered condition following prosocial video game exposure. However, no significant difference in P300 waves in the right-parietal area were found between the two types of games. These findings fail to support the idea that short-term exposure to a prosocial video game can influence subsequent behaviors, and does not support the GLM.

Funding

This study was supported by the National Social Science Foundation for Education of China [BBA140049].

Acknowledgments

This study was supported by the National Social Science Foundation for Education of China (BBA140049). We thank Dr. Qian Zhang and Dr. Yinkai Yang at Southwest University (SWU) for feedback of the final Manuscript.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.chb.2017.10.014.

References

Algina, J., & Keselman, H. J. (2003). Approximate confidence intervals for effect sizes. *Educational and Psychological Measurement, 63*, 537–553.

Anderson, C. A., & Bushman, B. J. (2002). Human aggression. *Annual Review of Psychology, 53*, 27–51.

Anderson, C. A., Deuser, W. E., & Deneve, K. M. (1995). Hot temperatures, hostile affect, hostile cognition, and arousal: Tests of a general-model of affective aggression. *Personality and Social Psychology Bulletin, 21*(5), 434–448.

Anderson, C. A., & Dill, K. E. (2000). Video games and aggressive thoughts, feelings, and behavior in the laboratory and in life. *Journal of Personality and Social Psychology, 78*(4), 772–790.

Anderson, C. A., Shibuya, A., Ihori, N., Swing, E. L., Bushman, B. J., Sakamoto, A., … Saleem, M. (2010). Violent video game effects on aggression, empathy, and prosocial behavior in eastern and western countries: A meta-analytic review. *Psychological Bulletin, 136*(2), 151–173.

Balconi, M., & Canavesio, Y. (2013). Prosocial attitudes and empathic behavior in emotional positive versus negative situations: Brain response (ERPs) and source localization (LORETA) analysis. *Cognitive Processing, 14*(1), 63–72.

Bartholow, B. D., Bushman, B. J., & Sestir, M. A. (2006). Chronic violent video game exposure and desensitization to violence: Behavioral and event-related brain potential data. *Journal of Experimental Social Psychology, 42*(4), 532–539.

Buckley, K. E., & Anderson, C. A. (2006). A theoretical model of the effects and consequences of playing video games. In P. Vorderer, & J. Bryant (Eds.), *Playing video games motives responses and consequences* (pp. 363–378). Mahwah, NJ: Lawrence Erlbaum.

Carlson, R. W., Aknin, L. B., & Liotti, M. (2015). When is giving an impulse? An ERP investigation of intuitive prosocial behavior. *Social Cognitive and Affective Neuroscience, 11*(7), 1121–1129.

Chiu Loke, I., Evans, A. D., & Lee, K. (2011). The neural correlates of reasoning about prosocial-helping decisions: An event-related brain potentials study. *Brain Research, 1369*, 140–148.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates.

Cox, A., Kohls, G., Naples, A. J., Mukerji, C. E., Coffman, M. C., Rutherford, H. J. V., … McPartland, J. C. (2015). Diminished social reward anticipation in the broad autism phenotype as revealed by event-related brain potentials. *Social Cognitive and Affective Neuroscience, 10*(10), 1357–1364.

Danziger, N., Prkachin, K. M., & Willer, J. C. (2006). Is pain the price of empathy? The perception of others' pain in patients with congenital insensitivity to pain. *Brain, 129*(9), 2494–2507.

Eisenberg, N., & Fabes, R. A. (1998). Prosocial development. In W. Damon (Ed.), *Handbook of child psychology: Social, emotional, and personality development: Vol. 3. t* (pp. 701–778). New York: Wiley.

Engelhardt, C. R., Bartholow, B. D., Kerr, G. T., & Bushman, B. J. (2011). This is your brain on violent video games: Neural desensitization to violence predicts increased aggression following violent video game exposure. *Journal of Experimental Social Psychology, 47*(5), 1033–1036.

Engen, H. G., & Singer, T. (2013). Empathy circuits. *Current Opinion in Neurobiology, 23*(2), 275–282.

Entertainment Software Association. (2013). *2013 Essential facts about the computer and video game industry*. Retrieved Feb 2013, from http://www.theesa.com/facts/pdfs/ESA_EF_2013.pdf.

Fan, Y., & Han, S. H. (2008). Temporal dynamic of neural mechanisms involved in empathy for pain: An event-related brain potential study. *Neuropsychologia, 46*(1), 160–173.
Ferguson, C. J. (2015). Do Angry Birds make for angry children? A meta-analysis of video game influences on children's and adolescents' aggression, mental health, prosocial behavior, and academic performance. *Perspectives on Psychological Science, 10*(5), 646–666.
Ferguson, C. J., & Dyck, D. (2012). Paradigm change in aggression research: The time has come to retire the general aggression model. *Aggression and Violent Behavior, 17*(3), 220–228.
Folstein, J. R., & Petten, C. V. (2008). Influence of cognitive control and mismatch on the N2 component of the ERP: A review. *Psychophysiology, 45*(1), 152–170.
Furuya-Kanamori, L., & Doi, S. A. (2016). Angry birds, angry children, and angry meta-analysts: A reanalysis. *Perspectives on Psychological Science, 11*(3), 408–414.
Gao, X., Pan, W., Li, C., Weng, L., Yao, M., & Chen, A. (2017). Long-time exposure to violent video games does not show desensitization on empathy for pain: An fMRI study. *Frontiers in Psychology, 8*(650). https://doi.org/10.3389/fpsyg.2017.00650.
Gentile, D. A., Anderson, C. A., Yukawa, S., Ihori, N., Saleem, M., Ming, L. K., … Sakamoto, A. (2009). The effects of prosocial video games on prosocial behaviors: International evidence from correlational, longitudinal, and experimental studies. *Personality and Social Psychology Bulletin, 35*(6), 752–763.
Graziano, W. G., Habashi, M. M., Sheese, B. E., & Tobin, R. A. (2007). Agreeableness, empathy, and helping: A person X situation perspective. *Journal of Personality and Social Psychology, 93*(4), 583–599.
Greitemeyer, T. (2011). Effects of prosocial media on social behavior when and why does media exposure affect helping and aggression? *Current Directions in Psychological Science, 20*(4), 251–255.
Greitemeyer, T., & Cox, C. (2013). There's no "I" in team: Effects of cooperative video games on cooperative behavior. *European Journal of Social Psychology, 43*(3), 224–228.
Greitemeyer, T., & Mügge, D. O. (2014). Video games do affect social outcomes: A meta-analytic review of the effects of violent and prosocial video game play. *Personality and Social Psychology Bulletin, 40*(5), 578–589.
Greitemeyer, T., & Osswald, S. (2010). Effects of prosocial video games on prosocial behavior. *Journal of Personality and Social Psychology, 98*(2), 211–221.
Greitemeyer, T., & Osswald, S. (2011). Playing prosocial video games increases the accessibility of prosocial thoughts. *Journal of Social Psychology, 151*(2), 121–128.
Greitemeyer, T., Osswald, S., & Brauer, M. (2010). Playing prosocial video games increases empathy and decreases schadenfreude. *Emotion, 10*(6), 796–802.
Hajcak, G., MacNamara, A., & Olvet, D. M. (2010). Event-related potentials, emotion, and emotion regulation: An integrative review. *Developmental Neuropsychology, 35*(2), 129–155.
Hajcak, G., Moser, J. S., Yeung, N., & Simons, R. F. (2005). On the ern and the significance of errors. *Psychophysiology, 42*(2), 151–160.
Hilgard, J., Engelhardt, C. R., & Rouder, J. N. (2017). Overstated evidence for short-term effects of violent games on affect and behavior: A reanalysis of Anderson et al. (2010). *Psychological Bulletin, 143*(7), 757–774.
Jerabeck, J. M., & Ferguson, C. J. (2013). The influence of solitary and cooperative violent video game play on aggressive and prosocial behavior. *Computers in Human Behavior, 29*(6), 2573–2578.
Kutner, L., & Olson, C. (2008). *Grand theft childhood: The surprising truth about violent video games and what parents can do*. New York, NY: Simon & Schuster.
Lahat, A., Helwig, C. C., & Zelazo, P. D. (2013). An event-related potential study of adolescents' and young adults' judgments of moral and social conventional violations. *Child Development, 84*(3), 955–969.
Liu, Y., Teng, Z., Lan, H., Zhang, X., & Yao, D. (2015). Short-term effects of prosocial video games on aggression: An event-related potential study. *Frontiers in Behavioral Neuroscience, 9*. https://doi.org/10.3389/Fnbeh.2015.00193.
Luck, S. J. (2014). *An introduction to the event-related potential technique*. Cambridge, MA: MIT Press.
Luo, P., Wang, J., Jin, Y., Huang, S., Xie, M., Deng, L., … Li, Y. (2014). Gender differences in affective sharing and self–other distinction during empathic neural responses to others' sadness. *Brain Imaging and Behavior, 9*(2), 312–322.
Morelli, S. A., Rameson, L. T., & Lieberman, M. D. (2014). The neural components of empathy: Predicting daily prosocial behavior. *Social Cognitive and Affective Neuroscience, 9*(1), 39–47.
Morey, R. D., Rouder, J. N., & Jamil, T. (2014). *BayesFactor: Computation of Bayes factors for common designs*. R package version 0.9, 8.
Nieuwenhuis, S., Aston-Jones, G., & Cohen, J. D. (2005). Decision making, the P3, and the locus coeruleus–norepinephrine system. *Psychological Bulletin, 131*(4), 510–532.
Pfabigan, D. M., Seidel, E.-M., Sladky, R., Hahn, A., Paul, K., Grahl, A., … Kranz, G. S. (2014). P300 amplitude variation is related to ventral striatum BOLD response during gain and loss anticipation: An EEG and fMRI experiment. *Neuroimage, 96*, 12–21.
Polich, J. (2007). Updating P300: An integrative theory of P3a and P3b. *Clinical Neurophysiology, 118*(10), 2128–2148.
Prot, S., Gentile, D. A., Anderson, C. A., Suzuki, K., Swing, E., Lim, K. M., … Lam, B. P. (2014). Long-term relations among prosocial-media use, empathy, and prosocial behavior. *Psychological Science, 25*(2), 358–368.
Przybylski, A. K., Deci, E. L., Rigby, C. S., & Ryan, R. M. (2014). Competence-impeding electronic games and players' aggressive feelings, thoughts, and behaviors. *Journal of Personality and Social Psychology, 106*(3), 441–457.
Read, G. L., Ballard, M., Emery, L. J., & Bazzini, D. G. (2016). Examining desensitization using facial electromyography: Violent videogames, gender, and affective responding. *Computers in Human Behavior, 62*(C), 201–211.
Rouder, J. N., Morey, R. D., Speckman, P. L., & Province, J. M. (2012). Default Bayes factors for ANOVA designs. *Journal of Mathematical Psychology, 56*(5), 356–374.
Sauer, J. D., Drummond, A., & Nova, N. (2015). Violent video games: The effects of narrative context and reward structure on in-game and postgame aggression. *Journal of Experimental Psychology Applied, 21*(3), 205–214.
Shamay-Tsoory, S. G., Aharon-Peretz, J., & Perry, D. (2009). Two systems for empathy: A double dissociation between emotional and cognitive empathy in inferior frontal gyrus versus ventromedial prefrontal lesions. *Brain, 132*(3), 617–627.
Singer, T., & Lamm, C. (2009). *The social neuroscience of empathy. Year in cognitive neuroscience 2009, 1156* pp. 81–96).
Szycik, G. R., Mohammadi, B., Hake, M., Kneer, J., Samii, A., Münte, T. F., et al. (2017a). Excessive users of violent video games do not show emotional desensitization: An fMRI study. *Brain Imaging and Behavior, 11*(3), 736–743.
Szycik, G. R., Mohammadi, B., Münte, T. F., & Te Wildt, B. T. (2017b). Lack of evidence that neural empathic responses are blunted in excessive users of violent video games: An fMRI study. *Frontiers in Psychology, 8*, 174. https://doi.org/10.3389/fpsyg.2017.00174.
Szűcs, D., & Soltész, F. (2010). Stimulus and response conflict in the color-word stroop task: A combined electro-myography and event-related potential study. *Brain Research, 1325*(1325), 63–76.
Tear, M. J., & Nielsen, M. (2013). Failure to demonstrate that playing violent video games diminishes prosocial behavior. *PLoS One, 8*(7), e68382.
Van Veen, V., & Carter, C. S. (2002). The anterior cingulate as a conflict monitor: fMRI and ERP studies. *Physiology & Behavior, 77*(4), 477–482.
Walter, H. (2012). Social cognitive neuroscience of Empathy: Concepts, circuits, and genes. *Emotion Review, 4*(1), 9–17.
Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect - the panas scales. *Journal of Personality and Social Psychology, 54*(6), 1063–1070.
Xiao, F., Zheng, Z., Wang, Y., Cui, J., & Chen, Y. (2015). Conflict monitoring and stimulus categorization processes involved in the prosocial attitude implicit association test: Evidence from event-related potentials. *Social Neuroscience, 10*(4), 408–417.
Yao, D. Z. (2001). A method to standardize a reference of scalp EEG recordings to a point at infinity. *Physiological Measurement, 22*(4), 693–711.
Yoder, K. J., & Decety, J. (2014). The good, the bad, and the just: Justice sensitivity predicts neural response during moral evaluation of actions performed by others. *Journal of Neuroscience, 34*(12), 4161–4166.